

Supplementary information for Adiabatic nonabelian braiding of imperfect Majoranas

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I. REALIZATION OF THE BRAIDING PROTOCOL VIA MINIMAL KITAEV CHAINS

Here we describe a possible realization of the generalized braiding protocol based on minimal Kitaev chains [1–7]. A single such system consists of two spin-polarized QDs ($l = 1, 2$) holding a single fermionic state each, that couple via a narrow superconductor

$$H = \sum_{l=1,2} \xi_l n_l + t_{\text{cot}} c_1^\dagger c_2 + \Delta_{\text{car}} c_1^\dagger c_2^\dagger + \text{H.c.} \quad (\text{S1})$$

Here, ξ_l are the single-particle energies of the QDs. t_{cot} and Δ_{car} are the elastic co-tunneling and crossed Andreev reflection amplitudes, which are tunable using electrostatic gates [6, 8, 9]. For $t_{\text{cot}} = \Delta_{\text{car}}$, $\xi_1, \xi_2 = 0$ this system has a degenerate ground state sector described by two MBSs $\gamma, \tilde{\gamma}$ each localized on one QD, see Ref. [3]. Deviations from $t_{\text{cot}} = \Delta_{\text{car}}$ lead to a splitting of the ground state expressed via a coupling ϵ of the two MBSs

$$H_c = \epsilon i\gamma\tilde{\gamma}, \quad \epsilon = \Delta_{\text{car}} - t_{\text{cot}}. \quad (\text{S2})$$

Next, we build a braiding setup by connecting three minimal Kitaev chains, as shown in Fig. 1(a) of the main paper. Each system hosts a pair of MBSs. The inner QDs of the three systems are connected via tunnel couplings

$$H = \sum_{k=1}^3 \epsilon_k i\gamma_k \tilde{\gamma}_k + t_{12} i\gamma_1 \gamma_2 + t_{13} i\gamma_1 \gamma_3, \quad (\text{S3})$$

where fluxes Φ_{12}, Φ_{13} through the superconducting loops have been used to make the tunnel couplings t_{12}, t_{13} real-valued.

The braiding protocol is defined by the Hamiltonian in Eq. (S3) where the time-dependent parameters $\epsilon_1(t), t_{12}(t), t_{13}(t)$ implement the coupling-based braiding protocol [10–16] while we keep $\epsilon_2, \epsilon_3 = 0$.

Next, we generalize the system by considering the possibility of spatially overlapping MBSs, meaning that the ‘outer’ MBSs $\tilde{\gamma}_k$ are partially located on the inner QD together with γ_k . This leads to a change in the tunnel coupling between the inner QDs, which is described by the transformation

$$\gamma_k \rightarrow \frac{1}{\sqrt{1+\zeta^2}} \gamma_k + i \frac{\zeta}{\sqrt{1+\zeta^2}} \tilde{\gamma}_k, \quad (\text{S4})$$

that interpolates between perfectly localized MBSs ($\zeta = 0$) and trivial states ($\zeta = 1$). For the two-site Kitaev

chain, this parameter directly connects to the MBS polarization M via $M = \frac{1-\zeta^2}{1+\zeta^2}$. In the case of minimal Kitaev chains, this introduces a dependence of ζ on the energy detuning ϵ_k . However, Ref. [15] shows that the dependence is minimal and a simultaneous gate tuning of $\xi_l, t_{\text{cot}}, \Delta_{\text{car}}$ enables a tuning of the coupling ϵ_k while keeping the MBS polarization, and therefore ζ , approximately constant. Therefore, in the analysis in the main paper, we assume that ζ is a constant system parameter independent of the variables ϵ_k .

The transformation in Eq. (S4) changes the system Hamiltonian as

$$H \rightarrow H + \tilde{H}, \quad (\text{S5})$$

which rescales the tunnel coupling t_{12}, t_{13} in H and introduces contributions via $\tilde{H}$ that stem from the fermionic character of the coupled operators

$$H = \epsilon_1 i\gamma_1 \tilde{\gamma}_1 + \frac{1}{1+\zeta^2} t_{12} i\gamma_1 \gamma_2 + \frac{1}{1+\zeta^2} t_{13} i\gamma_1 \gamma_3, \quad (\text{S6})$$

$$\tilde{H} = \frac{\zeta^2}{1+\zeta^2} t_{12} i\tilde{\gamma}_1 \tilde{\gamma}_2 + \frac{\zeta^2}{1+\zeta^2} t_{13} i\tilde{\gamma}_1 \tilde{\gamma}_3.$$

One of the main effects of $\tilde{H}$ is to lift the ground state degeneracy $\delta E \neq 0$ [15, 17]. If the protocol is adiabatic compared to δE , its outcome is trivial: the system remains in its original state. For a protocol that is diabatic with respect to δE , the splitting instead leads to a dynamical phase, which makes the result dependent on the exact strengths and durations of the time-dependent couplings.

The ground state degeneracy can be recovered via the following correction term

$$H_{\text{corr}} = \Lambda i\gamma_2 \tilde{\gamma}_2 + \Lambda i\gamma_3 \tilde{\gamma}_3, \quad (\text{S7})$$

which uses the two remaining MBS overlaps in the left and right MBS pairs, which for a minimal Kitaev chain can be tuned via t_{cot} and Δ_{car} . In general, $\Lambda(t)$ is a time-dependent parameter that has to be tuned along with the protocol. Choosing the same overlap for both systems is a possible, but not a necessary, choice that simplifies the analytical solution. This yields the full Hamiltonian.

$$H_{\text{tot}} = H + \tilde{H} + H_{\text{corr}}. \quad (\text{S8})$$

The energy splitting Δ between the degenerate ground state sector and the excited states is given by

$$\Delta = \sqrt{\epsilon_1^2 + \frac{1}{(1+\zeta^2)^2} (t_{12}^2 + t_{13}^2)}, \quad (\text{S9})$$

which we use to define the dimensionless parameters

$$\begin{aligned}\rho_1 &= \epsilon_1/\Delta, \quad \rho_{(k=2,3)} = \frac{1}{1+\zeta^2} t_{1k}/\Delta, \\ \lambda &= \Lambda/\Delta, \quad \eta = \zeta^2.\end{aligned}\quad (\text{S10})$$

We now arrive at the dimensionless Hamiltonian in Eq. (1) of the main paper

$$h = H_{\text{tot}}/\Delta = i\gamma_1\gamma_\Delta + \eta(\rho_2 i\tilde{\gamma}_1\tilde{\gamma}_2 + \rho_3 i\tilde{\gamma}_1\tilde{\gamma}_3) + \lambda(i\gamma_2\tilde{\gamma}_2 + i\gamma_3\tilde{\gamma}_3). \quad (\text{S11})$$

II. DIAGONALIZING THE HAMILTONIAN

In the following we will go through the diagonalization of Eq. (S11), which is the first step needed in the analytical solution for the result of the adiabatic braiding protocol. We start by introducing γ_Δ which is defined by

$$\gamma_\Delta = \cos\theta\tilde{\gamma}_1 + \sin\theta\cos\phi\gamma_2 + \sin\theta\sin\phi\gamma_3, \quad (\text{S12})$$

expressed in via the angles θ, ϕ defined as

$$\begin{aligned}\rho_1 &= \cos(\theta), \\ \rho_2 &= \sin(\theta)\cos(\phi), \\ \rho_3 &= \sin(\theta)\sin(\phi).\end{aligned}\quad (\text{S13})$$

This rotation of the $\tilde{\gamma}_1, \gamma_2, \gamma_3$ basis enforces the introduction of the following two rotated orthogonal MBSs

$$\begin{aligned}\gamma'_\theta &= -\sin\theta\tilde{\gamma}_1 + \cos\theta\cos\phi\gamma_2 + \cos\theta\sin\phi\gamma_3, \\ \gamma'_\phi &= -\sin\phi\gamma_2 + \cos\phi\gamma_3.\end{aligned}\quad (\text{S14})$$

In a similar way, we choose to rotate the MBSs $\tilde{\gamma}_2, \tilde{\gamma}_3$ as

$$\begin{aligned}\gamma_\eta &= \cos\phi\tilde{\gamma}_2 + \sin\phi\tilde{\gamma}_3, \\ \gamma'_\eta &= -\sin\phi\tilde{\gamma}_2 + \cos\phi\tilde{\gamma}_3.\end{aligned}\quad (\text{S15})$$

γ_1 is the only MBS that we do not rotate. Rewriting the Hamiltonian in this rotated basis yields

$$\begin{aligned}h &= i\gamma_1\gamma_\Delta + i\gamma_\Delta\gamma_\eta(\lambda\sin\theta + \eta\cos\theta\sin\theta) \\ &\quad + i\gamma_\eta\gamma'_\theta(\eta\sin^2\theta - \lambda\cos\theta) \\ &\quad + i\gamma'_\phi\gamma'_\eta\lambda.\end{aligned}\quad (\text{S16})$$

What we achieved by these rotations is to decouple the MBS pair $\gamma'_\phi, \gamma'_\eta$. Now we aim to diagonalize the remaining Hamiltonian by the ansatz

$$\begin{aligned}\gamma_1^D &= \alpha\gamma_1 + \beta\gamma_\eta, \quad \gamma_\eta^D = \alpha\gamma_\eta - \beta\gamma_1, \\ \gamma_\Delta^D &= \mu\gamma_\Delta + \nu\gamma'_\theta, \quad \gamma'_\theta^D = \mu\gamma'_\theta - \nu\gamma_\Delta.\end{aligned}\quad (\text{S17})$$

Inserting this ansatz in the Hamiltonian yields

$$\begin{aligned}h &= i\gamma_1^D\gamma_\Delta^D(\alpha\mu - \tilde{\lambda}\beta\mu + \tilde{\eta}\beta\nu) \\ &\quad + i\gamma_\eta^D\gamma'_\theta^D(\beta\nu + \tilde{\lambda}\alpha\nu + \tilde{\eta}\alpha\mu) \\ &\quad + i\gamma_\Delta^D\gamma_\eta^D(\mu\beta + \tilde{\lambda}\alpha\mu - \tilde{\nu}\alpha\nu) \\ &\quad + i\gamma_1^D\gamma'_\theta^D(-\alpha\nu + \tilde{\lambda}\beta\nu + \tilde{\nu}\beta\mu) \\ &\quad + i\gamma'_\phi\gamma'_\eta\lambda,\end{aligned}\quad (\text{S18})$$

where we defined

$$\begin{aligned}\tilde{\lambda} &= \lambda\sin\theta + \eta\cos(\theta)\sin(\theta), \\ \tilde{\eta} &= \eta\sin^2\theta - \lambda\cos\theta.\end{aligned}\quad (\text{S19})$$

Next, we demand that this new basis decouples the MBS pair $\gamma_1^D, \gamma_\Delta^D$ from the pair $\gamma_\eta^D, \gamma'_\theta^D$, which means that the terms proportional to $i\gamma_\Delta^D\gamma_\eta^D$ and $i\gamma_1^D\gamma'_\theta^D$ have to be zero

$$\begin{aligned}\mu\beta + \tilde{\lambda}\alpha\mu - \tilde{\nu}\alpha\nu &\stackrel{!}{=} 0, \\ -\alpha\nu + \tilde{\lambda}\beta\nu + \tilde{\nu}\beta\mu &\stackrel{!}{=} 0.\end{aligned}\quad (\text{S20})$$

This can be satisfied by introducing two angles $\theta_\alpha, \theta_\mu$ defined as

$$\begin{aligned}\theta_\mu &= -\frac{1}{2}\arctan\left(\frac{2\tilde{\lambda}\tilde{\eta}}{1 + \tilde{\lambda}^2 - \tilde{\eta}^2}\right), \\ \theta_\alpha &= -\arctan\left(-\tilde{\eta}\tan\theta_\mu + \tilde{\lambda}\right),\end{aligned}\quad (\text{S21})$$

which yield the previously introduced basis components

$$\begin{aligned}\alpha &= \cos\theta_\alpha, \quad \beta = \sin\theta_\alpha, \\ \mu &= \cos\theta_\mu, \quad \nu = \sin\theta_\mu,\end{aligned}\quad (\text{S22})$$

and allows writing the Hamiltonian in the fully diagonal form

$$h = \tilde{\Delta}i\gamma_1^D\gamma_\Delta^D + \varepsilon i\gamma_\eta^D\gamma'_\theta^D + \lambda i\gamma'_\phi\gamma'_\eta, \quad (\text{S23})$$

The Hamiltonian is defined such that the overall gap is given by

$$\tilde{\Delta} = \alpha\mu - \tilde{\lambda}\beta\mu + \tilde{\eta}\beta\nu, \quad (\text{S24})$$

that can be shown to be positive. This means that the low-energy sector is defined by

$$i\gamma_1^D\gamma_\Delta^D = -1, \quad (\text{S25})$$

which will stay constant throughout the protocol. Within this low-energy sector there are the additional smaller splittings λ and

$$\varepsilon = \beta\nu + \tilde{\lambda}\alpha\nu + \tilde{\eta}\alpha\mu = \tilde{\eta}\alpha/\mu, \quad (\text{S26})$$

coupled to the parities $i\gamma'_\phi\gamma'_\eta$ and $i\gamma_\eta^D\gamma'_\theta^D$. The last step in Eq. (S26) can be shown via Eq. (S20). In general, this means that the low-energy states are not degenerate. Finally, we determine how to tune λ to establish degeneracy. For this, we make use of the conservation of total parity. The basis definitions Eqs. (Eq. (S12)), (Eq. (S14)), (Eq. (S15)), and (Eq. (S17)) define how each MBS evolves under the implemented transformation:

$$\begin{aligned}\gamma_1 &\rightarrow \gamma_1^D, \quad \tilde{\gamma}_1 \rightarrow \gamma_\Delta^D, \\ \gamma_2 &\rightarrow \gamma'_\theta^D, \quad \tilde{\gamma}_2 \rightarrow \gamma_\eta^D, \\ \gamma_3 &\rightarrow \gamma'_\phi, \quad \tilde{\gamma}_3 \rightarrow \gamma'_\eta.\end{aligned}\quad (\text{S27})$$

This allows us to express the total parity $\sigma = \pm 1$ at any point of the protocol as

$$\begin{aligned}\sigma &= i\gamma_1\tilde{\gamma}_1 \cdot i\gamma_2\tilde{\gamma}_2 \cdot i\gamma_3\tilde{\gamma}_3 \\ &= i\gamma_1^D\gamma_\Delta^D \cdot i\gamma_\theta^D\gamma_\eta^D \cdot i\gamma_\phi'\gamma_\eta'.\end{aligned}\quad (\text{S28})$$

Inserting Eq. (S25) into Eq. (S28) yields

$$i\gamma_\phi'\gamma_\eta' = \sigma \cdot i\gamma_\eta^D\gamma_\theta'^D, \quad (\text{S29})$$

which we can reinsert into the Hamiltonian to find

$$h = \tilde{\Delta} i\gamma_1^D\gamma_\Delta^D + (\varepsilon + \sigma \cdot \lambda) i\gamma_\eta^D\gamma_\theta'^D. \quad (\text{S30})$$

Therefore, the necessary value of λ to obtain degeneracy in the ground state sector with total parity σ leads to Eq. (11) in the main paper:

$$\varepsilon(\eta, \lambda) + \sigma\lambda = 0. \quad (\text{S31})$$

In comparison to Eq. (11) of the main paper, Eq. (S31) is valid for both total parities σ . This equation looks simple, but proves to be very difficult to solve in the general case due to the dependence of ε on α, μ , which in turn depend on λ . We will show below that for the calculation of the Berry phase it is sufficient to solve this equation during the second protocol step defined by $\rho_1 = 0$. In this parameter limit it simplifies crucially.

III. PROTOCOL DEPENDENCE ON PARITY SECTOR AND SIGN CHOICES

Eq. (S31) shows a crucial difference in the braiding protocol performance between both total parities ($\sigma = \pm 1$). Evaluating Eq. (S31) for the fermionic limit ($\eta = 1$) at the beginning and end of the protocol $\theta \ll 1$ yields

$$\begin{aligned}\varepsilon(\theta, \lambda) &= -\lambda - \theta^2 \frac{1}{(-1 + \lambda)} + \mathcal{O}(\theta^3), \\ \Rightarrow 0 &= -\lambda - \theta^2 \frac{1}{(-1 + \lambda)} + \sigma\lambda + \mathcal{O}(\theta^3) \\ &= (-1 + \sigma)\lambda - \theta^2 \frac{1}{(-1 + \lambda)} + \mathcal{O}(\theta^3),\end{aligned}\quad (\text{S32})$$

which enforces the condition $\lambda = 0 + \mathcal{O}(\theta^2)$ for odd total parity ($\sigma = -1$). On the other hand, for even total parity ($\sigma = 1$), achieving degeneracy requires a big correction $\lambda = 1 + \mathcal{O}(\theta)$. As a consequence, the low-energy states are degenerate at energies

$$E_{\text{even, low}}/\Delta = -1 + \mathcal{O}(\theta), \quad (\text{S33})$$

but the high-energy states are strongly split into the energies

$$\begin{aligned}E_{\text{even, high}}/\Delta &= 1 \pm (\varepsilon - \lambda) \\ &= 1 \pm 2 + \mathcal{O}(\theta).\end{aligned}\quad (\text{S34})$$

Therefore, the correction pulls one of the high-energy states down to $E_{\text{even, high}} = -1$ and completely closes the gap to the low-energy sector for $\eta = 1$. For $\eta < 1$, the gap remains finite but is reduced compared to the $\sigma = -1$ case in the main paper.

There are therefore two distinct protocols, depending on the total parity. One with a reduced gap and one with a larger gap. Which parity sector that has the larger gap depends on the signs of the different parameters in Eq. (S11). To consider the effect of different signs, we check if a sign switch is equivalent to a change of basis of Majoranas, in which case it doesn't affect the resulting physics.

Consider first switching the sign of ρ_2 . This is equivalent to the transformation

$$\begin{aligned}\gamma_2 &\rightarrow -\gamma_2 \\ \tilde{\gamma}_2 &\rightarrow -\tilde{\gamma}_2,\end{aligned}\quad (\text{S35})$$

which doesn't change the physics. A similar argument applies to the signs of ρ_3 and η which also do not affect the protocol. The sign of λ is also unimportant since its value is defined by optimizing for degeneracy.

On the other hand, if ρ_1 is negative the protocol will work differently. Changing the sign of ρ_1 is equivalent to

$$\begin{aligned}\gamma_1 &\rightarrow -\gamma_1, \\ \gamma_2 &\rightarrow -\gamma_2, \\ \gamma_3 &\rightarrow -\gamma_3.\end{aligned}\quad (\text{S36})$$

Since an odd number of Majoranas changed sign, even and odd parity are exchanged. This version of the protocol has a larger gap in the even sector.

Another sign flip is attained with the transformation $\tilde{\gamma}_3 \rightarrow -\tilde{\gamma}_3$ which transforms the Hamiltonian as

$$\begin{aligned}h &\rightarrow i\gamma_1\gamma_\Delta + \eta(\rho_2 i\tilde{\gamma}_1\tilde{\gamma}_2 - \rho_3 i\tilde{\gamma}_1\tilde{\gamma}_3) \\ &\quad + \lambda(i\gamma_2\tilde{\gamma}_2 - i\gamma_3\tilde{\gamma}_3),\end{aligned}\quad (\text{S37})$$

and maps even to odd parity. In this form, the gap is larger in the even sector. Note that the error and correction term have the same relative signs between the two terms. If that is violated, the protocol fails to establish degeneracy between the ground states.

For different signs than in the main paper, some of the intermediate formulas in the derivation need to be modified, but the final result for the MBS similarity (Eq. (15) in the main paper) is the same as long as the error and correction terms have the same relative signs between their respective terms.

IV. DIAGONALIZING TWO COUPLED GROUPS OF MBSS

Analytically diagonalizing the Hamiltonian was possible because the Hamiltonian consists of two groups of

MBSs $\vec{\gamma}_L, \vec{\gamma}_R$ with only inter-group couplings. In this case, diagonalization does not mix the two groups, as we show in this section.

The Hamiltonian can be written as

$$H = i\vec{\gamma}_L^T M \vec{\gamma}_R, \quad (\text{S38})$$

where M is a real matrix. The singular value decomposition of M is $M = U^T S V$ where U and V are real orthogonal matrices and S is diagonal. Real orthogonal matrices preserve the canonical anticommutation relations, so

$$\vec{\chi}_L = U \vec{\gamma}_L, \quad (\text{S39})$$

$$\vec{\chi}_R = V \vec{\gamma}_R, \quad (\text{S40})$$

is a valid basis of MBSs. In this basis, the Hamiltonian takes the canonical diagonalized form

$$H = i\vec{\chi}_L^T S \vec{\chi}_R, \quad (\text{S41})$$

which doesn't mix the groups. During adiabatic time-evolution, these groups evolve separately.

V. CALCULATION OF THE BERRY PHASE

After we diagonalized the Hamiltonian and enforced degeneracy, we can solve the time evolution of the ground state manifold via the calculation of the Berry phase which is given by (Eq. (12) in the main paper)

$$U(\eta) = \mathcal{T} \exp \left\{ - \oint [P, \partial_\omega P] d\omega \right\}, \quad (\text{S42})$$

i.e., a time-ordered ($\mathcal{T}$) path integral over the full protocol parametrized by $\oint d\omega$. The projection operator onto the ground state manifold P is found by using the parity operator defined in Eq. (S25)

$$P = \frac{1}{2}(1 - i\gamma_1^D \gamma_\Delta^D). \quad (\text{S43})$$

The solution simplifies drastically if we split the path integral into the three protocol steps defined in the main paper

$$\oint d\omega = \int_{\pi/2}^0 d\theta + \int_0^{\pi/2} d\phi + \int_0^{\pi/2} d\theta. \quad (\text{S44})$$

We start by solving the first step of the protocol

$$U_{12} = \mathcal{T} \exp \left\{ - \int_0^{\pi/2} [P, \partial_\theta P] d\theta \right\}. \quad (\text{S45})$$

For this, we solve

$$\begin{aligned} \partial_\theta P &= -\frac{i}{2} \partial_\theta (\gamma_1^D \gamma_\Delta^D) \\ &= -\frac{i}{2} (\gamma_\eta^D \gamma_\Delta^D \cdot \partial_\theta \theta_\alpha + \gamma_1^D \gamma_\theta'^D \cdot (1 + \partial_\theta \theta_\mu)), \end{aligned} \quad (\text{S46})$$

where we made use of

$$\partial_\theta \gamma_1^D = \gamma_\eta^D \cdot \partial_\theta \theta_\alpha, \quad \partial_\theta \gamma_\Delta^D = \gamma_\theta'^D \cdot (1 + \partial_\theta \theta_\mu), \quad (\text{S47})$$

which can be checked via the Eqs. (Eq. (S12)), (Eq. (S14)), (Eq. (S15)), (Eq. (S17)), and (Eq. (S22)). This yields for the integrand

$$[P, \partial_\theta P] = \frac{1}{2} (\gamma_1^D \gamma_\eta^D \cdot \partial_\theta \theta_\alpha + \gamma_\Delta^D \gamma_\theta'^D \cdot (1 + \partial_\theta \theta_\mu)). \quad (\text{S48})$$

By applying

$$\gamma_1^D \gamma_\eta^D \stackrel{\phi=0}{=} \gamma_1 \tilde{\gamma}_2, \quad \gamma_\Delta^D \gamma_\theta'^D = \gamma_\Delta \gamma_\theta' \stackrel{\phi=0}{=} \tilde{\gamma}_1 \gamma_2, \quad (\text{S49})$$

we can further simplify the integrand to

$$[P, \partial_\theta P] = \frac{1}{2} (\gamma_1 \tilde{\gamma}_2 \cdot \partial_\theta \theta_\alpha + \tilde{\gamma}_1 \gamma_2 \cdot (1 + \partial_\theta \theta_\mu)). \quad (\text{S50})$$

This simplification was crucial because the integrand of Eq. (S45) now consists of two constant matrices which can be commuted as they are acting on different pairs of MBSs. As the integrand commutes with itself throughout the integration, we can neglect the time-ordering $\mathcal{T}$. We find

$$\begin{aligned} - \int_0^{\pi/2} [P, \partial_\theta P] d\theta &= \frac{1}{2} \tilde{\gamma}_2 \gamma_1 \int_0^{\pi/2} \partial_\theta \theta_\alpha d\theta \\ &\quad + \frac{1}{2} \gamma_2 \tilde{\gamma}_1 \int_0^{\pi/2} (1 + \partial_\theta \theta_\mu) d\theta, \\ \theta_{\alpha,\mu} \big|_{\theta=0}^{\theta=\pi/2} &= \tilde{\gamma}_2 \gamma_1 \theta_{\alpha,23}/2 + \gamma_2 \tilde{\gamma}_1 (\pi/4 + \theta_{\mu,23}/2). \end{aligned} \quad (\text{S51})$$

This solution reveals a particularly interesting fact about the diagonal basis Eq. (S17) parametrized by the angles $\theta_\alpha, \theta_\mu$. These are in general functions of the protocol parametrization θ, ϕ and vary throughout the protocol. However, $\theta_\alpha, \theta_\mu$ are constant during the second protocol step, which can be checked by considering Eq. (S31) in the limit $\theta = \pi/2$. This motivates the definition

$$\theta_{\alpha,23} = \theta_\alpha(\theta = \pi/2, \phi), \quad \theta_{\mu,23} = \theta_\mu(\theta = \pi/2, \phi), \quad (\text{S52})$$

describing these angles exactly during the time interval τ_{23} . Using this definition, we obtain the final result

$$U_{12} = \exp(\tilde{\gamma}_2 \gamma_1 \theta_{\alpha,23}/2) \exp(\gamma_2 \tilde{\gamma}_1 (\pi/4 + \theta_{\mu,23}/2)), \quad (\text{S53})$$

which depends on the previously defined $\theta_{\alpha,23}, \theta_{\mu,23}$ but not on the path they take during the protocol. This is a direct consequence of the path independence of the Berry phase.

This result also provides the Berry phase U_{31} for the third protocol step. We find it by exchanging $2 \rightarrow 3$

and applying the hermitian conjugate to invert the time ordering of the induced coupling

$$U_{31} = \exp(\gamma_1 \tilde{\gamma}_3 \theta_{\alpha,23}/2) \exp(\tilde{\gamma}_1 \gamma_3 (\pi/4 + \theta_{\mu,23}/2)). \quad (\text{S54})$$

Now the only thing left to solve is the second step of the protocol. As before we begin by calculating

$$\begin{aligned} \partial_\phi P &= -\frac{i}{2} \partial_\phi (\gamma_1^D \gamma_\Delta^D), \\ &= -\frac{i}{2} ((\partial_\phi \gamma_1^D) \gamma_\Delta^D + \gamma_1^D (\partial_\phi \gamma_\Delta^D)), \\ &= -\frac{i}{2} (\beta_{23} \gamma'_\eta \gamma_\Delta^D + \gamma_1^D \mu_{23} \gamma'_\phi), \end{aligned} \quad (\text{S55})$$

where we defined

$$\mu_{23} = \mu(\theta = \pi/2, \phi), \quad (\text{S56})$$

analogously to Eq. (S52). In the same way we define $\nu_{23}, \alpha_{23}, \beta_{23}$. This definition already implies that the basis components are constant during the second protocol step, which was already argued in Eq. (S52). This implies

$$\partial_\phi \mu = \partial_\phi \nu = \partial_\phi \alpha = \partial_\phi \beta = 0 \quad \text{at} \quad \theta = \pi/2, \quad (\text{S57})$$

which was used as

$$\begin{aligned} \partial_\phi \gamma_1^D &= \beta_{23} \partial_\phi \gamma'_\eta = \beta_{23} \gamma'_\eta, \\ \partial_\phi \gamma_\Delta^D &\stackrel{\theta=\pi/2}{=} \mu_{23} \partial_\phi \gamma_\Delta \stackrel{\theta=\pi/2}{=} \mu_{23} \gamma'_\phi, \end{aligned} \quad (\text{S58})$$

in the calculation of Eq. (S55). We can use this result to obtain

$$\begin{aligned} [P, \partial_\phi P] &= \frac{1}{2} (\beta \gamma_1^D \gamma'_\eta + \mu \gamma_\Delta^D \gamma'_\phi) \\ &= \frac{1}{2} (\mu^2 \gamma_2 \gamma_3 + \beta^2 \tilde{\gamma}_2 \tilde{\gamma}_3 + \alpha \beta \gamma_1 \gamma'_\eta - \mu \nu \tilde{\gamma}_1 \gamma'_\phi). \end{aligned} \quad (\text{S59})$$

The first two summands of this result are very similar to the integrand obtained in the previous protocol step in Eq. (S50). The difficulty with Eq. (S59) lies in the summands proportional to $\gamma_1 \gamma'_\eta$ and $\tilde{\gamma}_1 \gamma'_\phi$ because these are not constant operators. They evolve during the integration. As a consequence of the operators in the integral of

$$U_{23} = \mathcal{T} \exp \left\{ - \int_0^{\pi/2} [P, \partial_\phi P] d\phi \right\}, \quad (\text{S60})$$

not being constant, also the integrands do not commute with one another. Therefore, we cannot neglect the time ordering $\mathcal{T}$. To solve this expression, we will instead rewrite Eq. (S60) into matrix form and afterwards into a differential equation. We begin by inserting (Eq. (S14)),

(Eq. (S15)) and splitting Eq. (S60) into two sets of MBSs

$$\begin{aligned} U_{23} &= \\ &\mathcal{T} \exp \left\{ -\frac{1}{2} \int_0^{\pi/2} \mu^2 \gamma_2 \gamma_3 - \mu \nu \tilde{\gamma}_1 (-\sin \phi \gamma_2 + \cos \phi \gamma_3) d\phi \right\} \\ &\cdot \mathcal{T} \exp \left\{ -\frac{1}{2} \int_0^{\pi/2} \beta^2 \tilde{\gamma}_2 \tilde{\gamma}_3 + \alpha \beta \gamma_1 (-\sin \phi \tilde{\gamma}_2 + \cos \phi \tilde{\gamma}_3) d\phi \right\}, \end{aligned} \quad (\text{S61})$$

which can be solved separately, and we will restrict the presented solution to the first line. This is possible because combinations of $\tilde{\gamma}_1, \gamma_2, \gamma_3$ commute with combinations of $\gamma_1, \tilde{\gamma}_2, \tilde{\gamma}_3$.

$$\begin{aligned} U_{23}^{\mu\nu} &= \\ &\mathcal{T} \exp \left\{ -\frac{1}{2} \int_0^{\pi/2} \mu^2 \gamma_2 \gamma_3 - \mu \nu \tilde{\gamma}_1 (-\sin \phi \gamma_2 + \cos \phi \gamma_3) d\phi \right\}. \end{aligned} \quad (\text{S62})$$

Next, we choose a Pauli matrix representation

$$\gamma_3 \tilde{\gamma}_1 = i\sigma_x, \quad \gamma_2 \tilde{\gamma}_1 = -i\sigma_y, \quad \Rightarrow i\sigma_z = \sigma_x \sigma_y = \gamma_2 \gamma_3, \quad (\text{S63})$$

which we insert into Eq. (S62)

$$U_{23}^{\mu\nu} = \mathcal{T} \exp \left\{ \frac{i}{2} \int_0^{\pi/2} -\mu^2 \sigma_z - \mu \nu (\sin \phi \sigma_y + \cos \phi \sigma_x) d\phi \right\}. \quad (\text{S64})$$

Next, we reexpress Eq. (S64) as the solution to a differential equation

$$\partial_\omega U_{23}^{\mu\nu}(\omega) = A(\omega) U_{23}^{\mu\nu}(\omega), \quad (\text{S65})$$

evaluated at

$$U_{23}^{\mu\nu} = U_{23}^{\mu\nu}(\omega = \pi/2), \quad (\text{S66})$$

with the initial condition

$$U_{23}^{\mu\nu}(\omega = 0) = \mathbb{1}_2, \quad (\text{S67})$$

where we defined

$$A(\omega) = \frac{i}{2} (-\mu^2 \sigma_z - \mu \nu (\sin \omega \sigma_y + \cos \omega \sigma_x)). \quad (\text{S68})$$

By solving the differential equation, we find

$$U_{23}^{\mu\nu} = \exp\{-i\pi/4\sigma_z\} \exp\{-i\pi/4\nu(\mu\sigma_y - \nu\sigma_z)\}, \quad (\text{S69})$$

which we rewrite back into MBSs

$$U_{23}^{\mu\nu} = \exp\{\pi/4\gamma_3\gamma_2\} \exp\{-\pi/4\nu(\mu\gamma_3\tilde{\gamma}_1 + \nu\gamma_3\gamma_2)\}. \quad (\text{S70})$$

Together with the second contribution $U_{23}^{\alpha\beta}$ solved in the same way we find

$$U_{23} = \exp\{\pi/4 \gamma_3 \gamma_2\} \exp\{\pi/4 \nu(\mu\gamma_3\tilde{\gamma}_1 + \nu\gamma_3\gamma_2)\} \cdot \exp\{\pi/4 \tilde{\gamma}_3\tilde{\gamma}_2\} \exp\{\pi/4 \alpha(\beta\tilde{\gamma}_3\gamma_1 - \alpha\tilde{\gamma}_3\tilde{\gamma}_2)\}. \quad (\text{S71})$$

The last remaining step is to multiply all contributions to obtain

$$U(\eta) = U_{31} U_{23} U_{12} = \exp\{\pi/4(1 + \nu_{23})\gamma_3\gamma_2\} \exp\{\pi/4(1 - \alpha_{23})\tilde{\gamma}_3\tilde{\gamma}_2\}, \quad (\text{S72})$$

where we used the identities

$$\begin{aligned} \mu_{23} &= \cos(\theta_{\mu,23}), & \nu_{23} &= \sin(\theta_{\mu,23}), \\ \alpha_{23} &= \cos(\theta_{\alpha,23}), & \beta_{23} &= \sin(\theta_{\alpha,23}). \end{aligned} \quad (\text{S73})$$

From this follows that a double braid implements the rotation

VI. SOLUTION OF THE DEGENERACY CONDITION IN THE SECOND PROTOCOL INTERVAL

The general solution of Eq. (S72) includes the parameters ν_{23} , α_{23} which are obtained by solving the degeneracy condition Eq. (S31) in protocol interval τ_{23} characterized by $\theta = \pi/2$, where Eq. (S31) simplifies to

$$\frac{\alpha_{23}}{\mu_{23}} = -\sigma \frac{\lambda_{23}}{\eta}, \quad (\text{S74})$$

which together with

$$\begin{aligned} \frac{\nu_{23}}{\mu_{23}} &= -\tan\left(\frac{1}{2} \arctan\left(\frac{2\lambda_{23}\eta}{1 + \lambda_{23}^2 - \eta^2}\right)\right), \\ \frac{\beta_{23}}{\alpha_{23}} &= -\tan\left(\arctan\left(-\eta \frac{\nu_{23}}{\mu_{23}} + \lambda_{23}\right)\right), \end{aligned} \quad (\text{S75})$$

obtained from Eq. (S21) yields the solution

$$\begin{aligned} \lambda_{23} &= -\sigma \frac{\eta}{\sqrt{1 + \eta^2}}, \\ \nu_{23} &= \frac{\sigma\eta^2}{\sqrt{1 + \eta^2 + \eta^4}}, \\ \alpha_{23} &= \frac{1}{\sqrt{1 + \eta^2 + \eta^4}}, \\ \mu_{23} &= \frac{\sqrt{1 + \eta^2}}{\sqrt{1 + \eta^2 + \eta^4}}, \\ \beta_{23} &= \frac{\sigma\eta\sqrt{1 + \eta^2}}{\sqrt{1 + \eta^2 + \eta^4}}. \end{aligned} \quad (\text{S76})$$

Eqs. (Eq. (S72)), (Eq. (S76)) together yield the solution for the angles

$$\begin{aligned} \phi &= \frac{\pi}{4} (1 + \nu_{23}) = \frac{\pi}{4} \left(1 + \frac{\sigma\eta^2}{\sqrt{1 + \eta^2 + \eta^4}}\right), \\ \tilde{\phi} &= \frac{\pi}{4} (1 - \alpha_{23}) = \frac{\pi}{4} \left(1 - \frac{1}{\sqrt{1 + \eta^2 + \eta^4}}\right), \end{aligned} \quad (\text{S77})$$

presented in Eq. (13) of the main paper.

VII. FUNCTIONAL DEPENDENCE BETWEEN PARITY MEASUREMENTS AND MBS SIMILARITY

In the following, we derive the functional connection between the MBS similarity in Eq. (15) of the main paper and a parity measurement.

We start by evaluating the MBS similarity as

$$\begin{aligned} S(U) &= \text{Tr}\{U^2(\eta)^\dagger U^2(0)\}^2/4, \\ &= \text{Tr}\left\{\exp\left(-2(\phi - \tilde{\phi})\gamma_3\gamma_2\right) \exp\left(2\frac{\pi}{4}\gamma_3\gamma_2\right)\right\}^2/4, \\ &= \text{Tr}\left\{\exp\left(2\left(\frac{\pi}{4} - \phi + \tilde{\phi}\right)\gamma_3\gamma_2\right)\right\}^2/4, \\ &= \text{Tr}\left\{\cos\left(2\left(\frac{\pi}{4} - \phi + \tilde{\phi}\right)\right) + \sin\left(2\left(\frac{\pi}{4} - \phi + \tilde{\phi}\right)\right)\gamma_3\gamma_2\right\}^2/4, \\ &= \cos^2\left(2\left(\frac{\pi}{4} - \phi + \tilde{\phi}\right)\right), \\ &= \sin^2\left(2\left(\phi - \tilde{\phi}\right)\right). \end{aligned} \quad (\text{S78})$$

Inserting $\phi, \tilde{\phi}$, as defined in Eq. (S77), followed by further analytical simplification yields the solution presented in Eq. (15) of the main paper.

For the parity measurement, we assume that the system is initialized in a state $|\psi\rangle$ by initializing the left and right parities within the ground state sector of the total odd total parity $\sigma = -1$

$$i\gamma_2\tilde{\gamma}_2|\psi\rangle = i\gamma_3\tilde{\gamma}_3|\psi\rangle = \pm|\psi\rangle. \quad (\text{S79})$$

We start by evaluating the protocol-induced transformation of the parities

$$\begin{aligned} U^2(\eta) i\gamma_2\tilde{\gamma}_2 U^2(\eta)^\dagger &= i \exp(2\phi\gamma_3\gamma_2)\gamma_2 \exp(-2\phi\gamma_3\gamma_2) \\ &\quad \exp\left(2\tilde{\phi}\tilde{\gamma}_3\tilde{\gamma}_2\right)\tilde{\gamma}_2 \exp\left(-2\tilde{\phi}\tilde{\gamma}_3\tilde{\gamma}_2\right), \\ &= i(\cos(4\phi)\gamma_2 + \sin(4\phi)\gamma_3)(\cos(4\tilde{\phi})\tilde{\gamma}_2 + \sin(4\tilde{\phi})\tilde{\gamma}_3). \end{aligned} \quad (\text{S80})$$

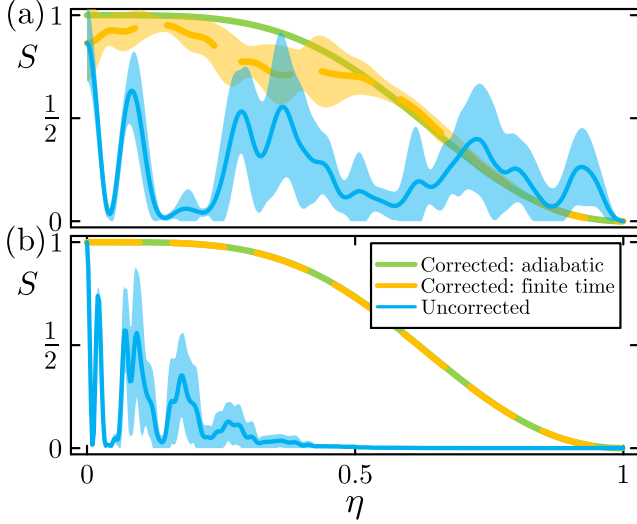


FIG. S1: Comparison between finite time numerical simulations of the corrected and uncorrected protocol, and the adiabatic result for the corrected protocol. We show the mean and one standard deviations calculated over a range of protocol times. (a) Short protocol times with $T\Delta = 60 \pm 12, k = 10$. (b) Longer protocol times with $T\Delta = 200 \pm 40, k = 10$.

Subsequent measurement of $i\gamma_2\tilde{\gamma}_2$ and $i\gamma_3\tilde{\gamma}_3$ yield

$$\begin{aligned}
 & \langle \psi | U^2(\eta) i\gamma_2\tilde{\gamma}_2 U^2(\eta)^\dagger | \psi \rangle \\
 &= \langle \psi | (\cos(4\phi)\gamma_2 + \sin(4\phi)\gamma_3) \\
 & \quad (\cos(4\tilde{\phi})\tilde{\gamma}_2 + \sin(4\tilde{\phi})\tilde{\gamma}_3) | \psi \rangle, \\
 &= \pm (\cos(4\phi)\cos(4\tilde{\phi})i\gamma_2 + \sin(4\phi)\sin(4\tilde{\phi})i\gamma_3), \\
 &= \pm (1 - 2\sin^2(2(\phi - \tilde{\phi}))), \\
 &= \pm (1 - 2S(U)),
 \end{aligned} \tag{S81}$$

where we used that $\langle \psi | i\gamma_2\tilde{\gamma}_3 | \psi \rangle = \langle \psi | i\gamma_3\tilde{\gamma}_2 | \psi \rangle = 0$.

VIII. NUMERICAL SIMULATIONS

To crosscheck our results, investigate diabatic effects and compare with the uncorrected protocol, we solve the time-dependent Schrödinger equation for the full unitary time evolution operator $U(\eta, t)$,

$$\begin{aligned}
 \partial_t U(\eta, t) &= iH(t)U(\eta, t), \\
 U(\eta, 0) &= \mathbb{1},
 \end{aligned} \tag{S82}$$

with $H(t)$ given by Eq. (S11) (Eq. (1) of the main paper). The couplings ρ_i are turned on and off periodically according to the function

$$f(t) = 1/2 + \tanh(k \cos(2\pi t/T))/2, \tag{S83}$$

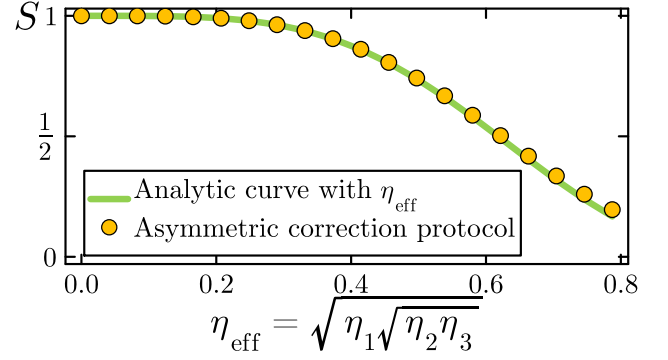


FIG. S2: Protocol with asymmetric MBS overlaps $\eta_2 = \eta_1/2, \eta_3 = \eta_1/5$. The markers show the MBS Similarity for numerical solutions with $T\Delta = 2000, k = 10$. The analytic curve is obtained by using η_{eff} in the analytical formulas.

where k determines how sharp the step is and T is the time for a single braid. The different couplings are obtained by shifting the smooth step function as

$$\begin{aligned}
 \rho_1 &= f(t/T)/N, \\
 \rho_2 &= f(t/T - 1/3)/N, \\
 \rho_3 &= f(t/T - 2/3)/N,
 \end{aligned} \tag{S84}$$

where the normalization factor N is there to ensure that $\sum_i \rho_i^2 = 1$. The precise parametrization is unimportant as long as it is adiabatic with respect to the large gap ($T\Delta \gg k$) and diabatic with respect to the smallest gap ($T\Delta e^{-2k} \ll k$). We set $k = 10$ in all our simulations. The parameter λ is chosen at each point in time so that the ground state sector is degenerate in the appropriate parity sector. This is done by solving Eq. (S31) with a numerical root-finding algorithm at each time step. The code we used for these calculations can be found at [18].

Figure S1 shows a comparison between numerical results for the MBS similarity and the analytical result of the main paper (Eq. (15)). For the numerical solutions, we show the mean and one standard deviation by averaging over a range of protocol times T . The top figure (a) is for a short protocol time with large diabatic errors. T is sampled uniformly in an interval where $T\Delta = 60 \pm 12$. The bottom figure (b) uses a longer protocol time with $T\Delta = 200 \pm 40$. The corrected protocol has no errors from a dynamical phase, and in this case very small diabatic errors.

IX. ASYMMETRIC MBS OVERLAPS

In the main paper, we have assumed that the MBS overlaps are symmetric, i.e., $\eta_1 = \eta_2 = \eta_3$. In this section, we consider asymmetric errors and find empirically that a suitably generalized version of the analytical results works very well even in this case. The Hamiltonian

is now

$$\begin{aligned} H/\Delta = & \rho_1 i\gamma_1 \tilde{\gamma}_1 + \rho_2 i\gamma_1 \gamma_2 + \rho_3 i\gamma_1 \gamma_3 \\ & + \rho_2 \sqrt{\eta_1 \eta_2} i\tilde{\gamma}_1 \tilde{\gamma}_2 + \rho_3 \sqrt{\eta_1 \eta_3} i\tilde{\gamma}_1 \tilde{\gamma}_3 \\ & + \lambda_2 i\gamma_2 \tilde{\gamma}_2 + \lambda_3 i\gamma_3 \tilde{\gamma}_3, \end{aligned} \quad (\text{S85})$$

and the correction term now includes two parameters: λ_2 and λ_3 . At each step in time, we numerically optimize these for energy degeneracy of the ground states.

We generalize the analytical result to account for the three different overlaps η_1, η_2, η_3 by defining an effective MBS overlap

$$\eta_{\text{eff}} = \sqrt{\eta_1 \sqrt{\eta_2 \eta_3}}, \quad (\text{S86})$$

which we use in place of η in the analytical result. This formula was obtained by testing different ways of taking the mean of the overlaps and this one worked the best. [Figure S2](#) shows a comparison between a numerical solution and the analytic curve using η_{eff} . We have plotted a range of η_{eff} where the numerical optimization for λ_2 and λ_3 was successful, and the curves agree very well. For larger values of η_{eff} , the optimization failed.

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